Assignment 1 AIML Lab (B4)

1. Write a Python program to add two numbers.

```
a = 10
b = 20
sum = a + b
print("Sum =", sum)
```

Output :

```
Sum = 30
```

2. Write a Python program to declare variables and display the data type of each variable.

```
a = 10
b = 10.5
c = "Hello"
d = True

print(type(a))
print(type(b))
print(type(c))
print(type(d))
```

Output :

```
<class 'int'>
<class 'float'>
<class 'str'>
<class 'bool'>
```

3. Write a Python program to demonstrate type casting in Python.

```
a = "10"
b = int(a)

c = 5
d = float(c)

print(b)
print(d)
```

Output:

```
10
5.0
```

4. Write a Python program to demonstrate logical operators .

```
a = 10
b = 5

print(a > 5 and b < 10)
print(a > 15 or b < 10)
print(not(a > 5))
```

Output:
True
True
False

5. Write a Python program to calculate the length of a string.

```
text = "Python"

print("Length =", len(text))
```

Output:
Length = 6

6. Write a Python program to create a new string using the first two and the last two characters of a given string.

```
text = "Python"

new_string = text[:2] + text[-2:]

print(new_string)
```

Output:
Pyon

7. Write a Python program to concatenate two strings.

```
a = "Hello"
b = "Python"

c = a + " " + b

print(c)
```

Output:
Hello Python

8. Write a Python program to find the greatest of three numbers.

```
a = 10
b = 25
c = 15

if a > b and a > c:
   print(a)
elif b > c:
   print(b)
else:
   print(c)
```

Output:
25

9. Write a Python program to check whether a given character is an uppercase alphabet or a lowercase alphabet.

```
ch = 'A'

if ch.isupper():
   print("Uppercase")
elif ch.islower():
   print("Lowercase")
else:
   print("Not an alphabet")
```

Output:
Uppercase

10. Write a Python program to determine whether the given input is a number or a character.

```
ch = '5'

if ch.isdigit():
   print("Number")
else:
   print("Character")
```

Output:
Number

11. Write a Python program to print the Fibonacci series from 0 to 50.

```
a = 0
b = 1

while a <= 50:
   print(a, end=" ")
   a, b = b, a + b
```

Output:
0 1 1 2 3 5 8 13 21 34

12. Write a Python program to remove the characters at odd index positions from a given string.

```
text = "Python"

new_string = text[::2]

print(new_string)
```

Output:
Pto

13. Frequency of each character in a string [EXTRA]

```
text = "hello"

for ch in set(text):
   print(ch, "=", text.count(ch))
```

Output:

```
h = 1
e = 1
l = 2
o = 1
```

14. Python program to display first non repeating character in a string [EXTRA]

```
text = "swiss"

for ch in text:
   if text.count(ch) == 1:
      print("First non-repeating character =", ch)
      break
```

Output:

```
First non-repeating character = w
```

15. Program to display minimum currency to pay the bill amount. (including 0.50) [EXTRA]

```
amount = 387.00

currency = [500, 200, 100, 50, 20, 10, 5, 2, 1, 0.50]

for coin in currency:
   count = int(amount // coin)
   amount = round(amount - count * coin, 2)
   print(coin, "=", count)
```

Output :

```
500 = 0
200 = 1
100 = 1
50 = 1
20 = 1
10 = 1
5 = 1
2 = 1
1 = 0
0.5 = 0
```